

HR Employee Attrition Analysis

Predicting and Preventing Employee Turnover | Elevate Labs Internship Project | 2026

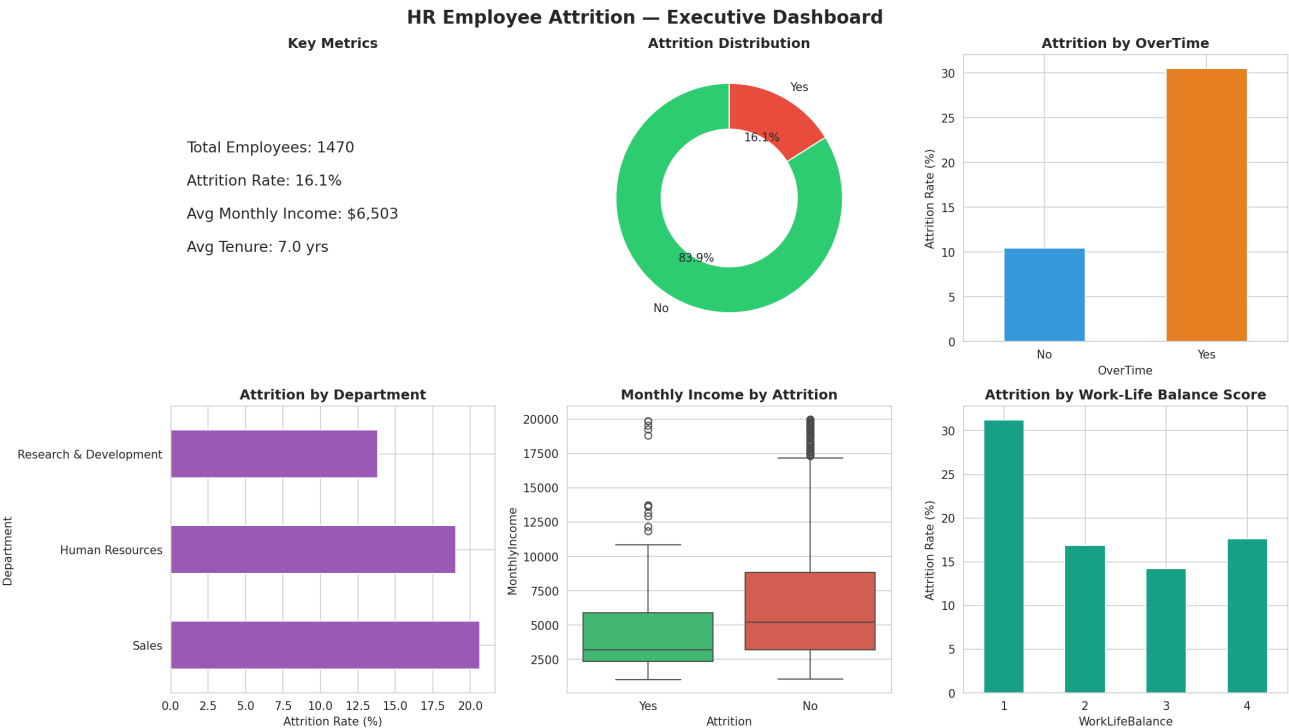
EMPLOYEES	ATTRITION RATE	BEST MODEL AUC	FLIGHT-RISK PAYROLL
1470	16.1%	0.808	\$2.10M

Dataset: IBM HR Analytics Employee Attrition (Kaggle) | **1470 employees**, 35 original attributes, zero missing values | **Tools:** Python, Pandas, Scikit-learn, Matplotlib/Seaborn, ReportLab

Introduction & Abstract

This project analyzes IBM's HR Employee Attrition dataset to identify the drivers behind voluntary employee turnover and to build a predictive model that flags employees at high risk of leaving. Of 1470 employees, 16.1% left the company, a class imbalance handled using class-weighted models rather than oversampling. The strongest single driver of attrition is overtime work: employees who work overtime leave at 30.5% versus 10.4% for those who don't — roughly a 3x gap. The analysis combines exploratory data analysis, feature engineering, a 4-model comparison, permutation-importance explainability, and a 3-tier risk segmentation that converts model output into employee-level, actionable retention targets.

Exploratory Data Analysis

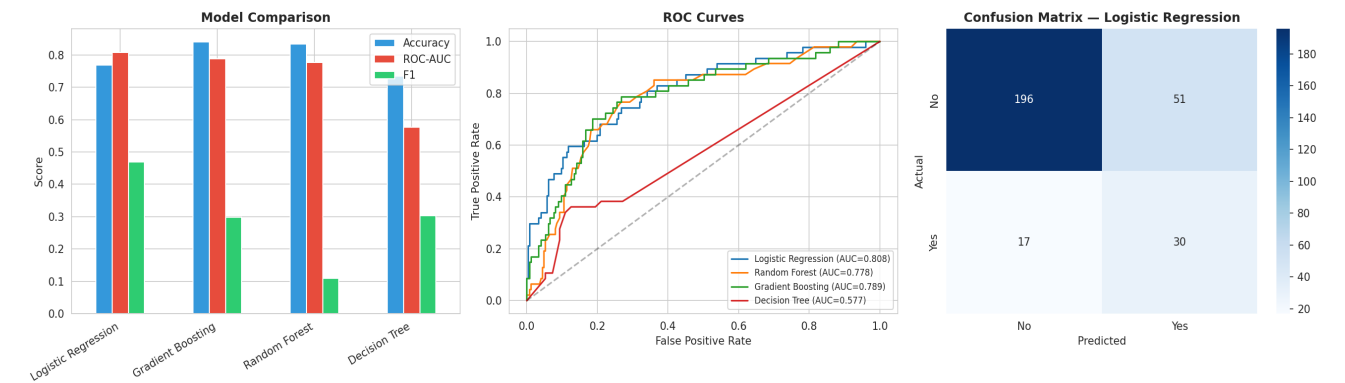


Key findings: Overtime is the single largest attrition driver (30.5% vs 10.4%). Sales has the highest departmental attrition (20.6%), led by Sales Representatives (39.8%). Employees who left earned \$2046 less per month on average than those who stayed. Single employees (25.5%) and those with low work-life balance scores (31.2%) churn at markedly higher rates than their peers.

Preprocessing & Feature Engineering

Three constant columns (EmployeeCount, Over18, StandardHours) and the EmployeeNumber ID column were dropped. Binary fields (Attrition, Gender, OverTime) were label-encoded; five multi-class categorical fields (BusinessTravel, Department, EducationField, JobRole, MaritalStatus) were one-hot encoded. Five engineered features were added — IncomePerYear, TenureRatio, AvgSatisfaction, PromotionGap, and IsYoungAndNew — bringing the feature count to 49. Continuous features were standard-scaled after an 80/20 stratified train-test split. Because attrition is imbalanced, all models used class_weight='balanced' rather than synthetic oversampling, so the loss function penalizes missed leavers more heavily.

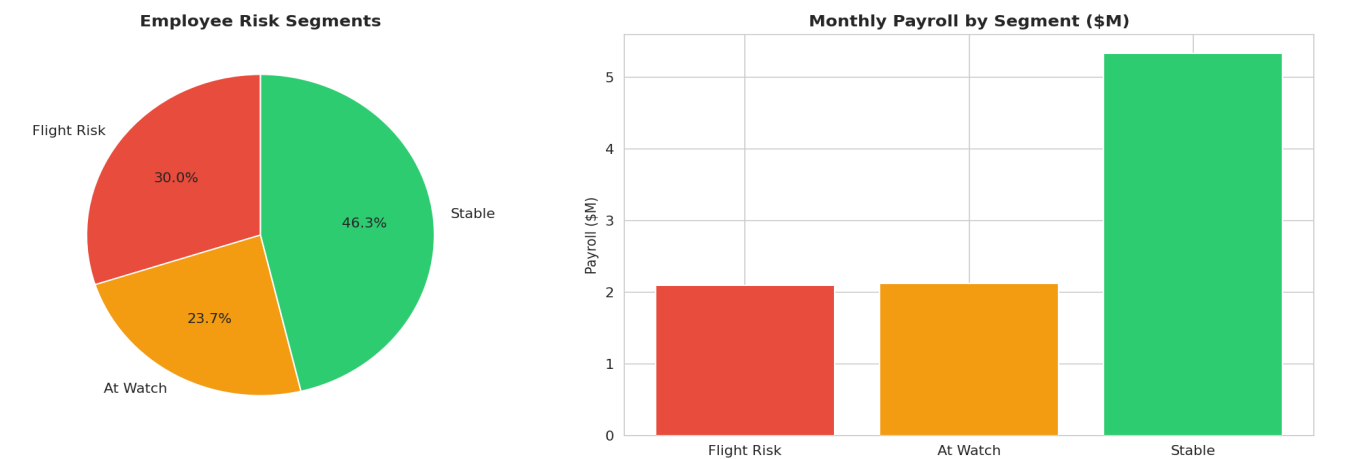
Model Building & Explainability



Model	Accuracy	ROC-AUC	F1 Score
Logistic Regression	76.9%	0.808	0.469
Gradient Boosting	84.0%	0.789	0.299
Random Forest	83.3%	0.778	0.109
Decision Tree	73.5%	0.577	0.304

Logistic Regression achieved the best ranking ability (ROC-AUC 0.808), making it the most reliable model for prioritizing which employees to focus retention efforts on. Permutation importance (a model-agnostic substitute for SHAP) confirms OverTime, frequent business travel, and number of prior companies worked as the top predictive signals.

Employee Risk Segmentation & Recommendations



Each employee was scored with the Logistic Regression model and segmented into three tiers: **Flight Risk** (441 employees, 40.6% actual attrition, \$2.10M monthly payroll), **At Watch** (349 employees, 10.9% actual attrition), and **Stable** (680 employees, 2.9% actual attrition). **Recommendations:** (1) Cap or compensate overtime for Sales roles, the highest-risk segment; (2) introduce structured promotion check-ins for employees flagged by low PromotionGap; (3) review compensation for employees earning below the \$2046/month retention gap; (4) extend stock option eligibility, since 0-stock-option employees churn at the highest rate; (5) prioritize the Flight Risk segment for stay interviews this quarter.

Conclusion: Attrition at this company is driven primarily by overtime burden, role/department (Sales), compensation gaps, and low work-life balance — all addressable through targeted HR policy rather than broad pay raises. The Logistic Regression model (AUC 0.808) combined with the 3-tier segmentation gives HR a ranked, actionable list of employees to engage with first.